

PURCHASE ORDER MANAGEMENT SYSTEM

Complete Project Documentation - Updated with BSC Exclusive Furniture & Lifestyle Catalogue

Functional Requirements Specification (FRS) + System Design + Workflows + Data Model + UI/UX + Testing Blueprint
+ BSC Merchandise Catalogue

System Organizational & Product Hierarchy

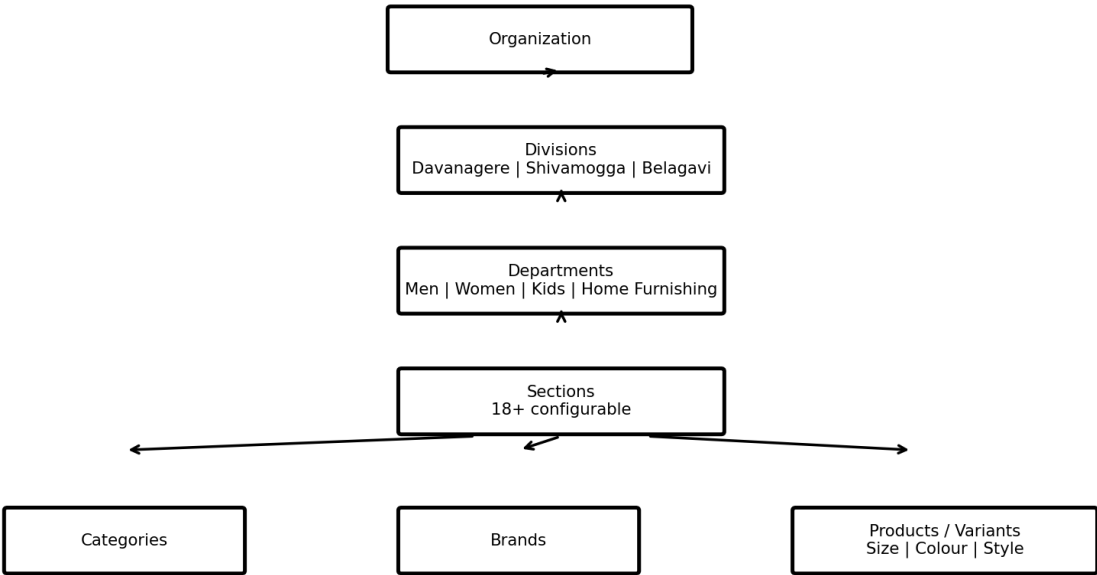


Figure 1. High-level organizational and product hierarchy.

Item	Value
Project Type	Full-stack enterprise web application
Primary Business	Retail merchandise procurement / purchase ordering
Divisions	Davanagere, Shivamogga, Belagavi
Initial Sections	Approximately 18, configuration-driven
Primary Roles	Super Admin, Admin/Domain Admin, Purchase User/Executive, Approver, Receiving User, Viewer, Auditor
Document Version	2.0
Prepared	5 September 2026
Source Basis	Uploaded Purchase Order Management System requirements and supplied master prompt

Document Control

Version	Date	Author / Owner	Description
1.0	5 Sep 2026	Project Team	Initial functional requirements
2.0	5 Sep 2026	Project Team	Expanded complete project documentation with diagrams, data model, architecture, UI, testing, security, deployment and implementation roadmap

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1. Executive Summary

This project is a centralized, secure and configurable Purchase Order (PO) and Merchandise Procurement Management System for a multi-division retail business. The application is designed to replace manual or spreadsheet-based purchasing with a controlled digital workflow that captures the complete purchasing decision: division, department, section, brand, product, product serial number, colour, size-wise quantity, purchase price, margin, discount, dealer and final value. The uploaded requirements define Davanagere, Shivamogga and Belagavi as the initial divisions and approximately 18 configurable merchandise sections.

The system is intentionally configuration-driven. Administrators must be able to create, modify, activate, deactivate, reorder and archive sections, sizes, colours, brands, products, dealers, approval rules and other master data without requiring code changes. Historical transactions must remain intact when a master record is retired.

The target operating model is a full procurement lifecycle: authenticate the user, select an authorized division, choose the department/section, choose or create master data, enter variant quantities, calculate commercial values, select a dealer, submit for approval, issue a purchase order, receive goods in partial or full quantities, update inventory, and expose the result through dashboards, reports and a full audit history.

2. Project Objectives and Success Criteria

- Digitize the complete purchase-order raising and approval process across all divisions and product sections.
- Provide a reusable master-data model instead of hard-coded categories.
- Capture apparel size curves and colour matrices efficiently for large purchase orders.
- Support special structures for sarees, jewellery, kids, footwear, accessories and home furnishing.
- Ensure brand number, brand name, brand serial number and product serial number are independently maintained.
- Allow catalogue-driven selection and safe manual entry through the Other option.
- Automatically calculate quantity, gross value, margin, discount, tax, charges and final value using server-side rules.
- Give Super Admin enterprise-wide visibility while restricting Domain Admins and other users to their authorized scope.
- Provide approval governance, amendment/versioning, receiving, inventory integration and auditability.
- Provide exportable operational and management reporting.
- Create a platform that can later integrate with accounting, ERP, POS, WMS, vendor portals and smart replenishment engines.

2.1 Success Criteria

- No unauthorized user can access another division's protected data.
- No approved PO can be silently altered.
- No master record duplication is permitted where the business identifier is unique.
- All key monetary calculations are reproducible and server-validated.
- PO history remains understandable after section/product/master-data changes.
- Users can enter size/colour quantities quickly without excessive navigation.
- Super Admin can consolidate all three divisions from one dashboard.
- Every critical change is attributable to a user, timestamp and before/after state.

3. Scope and Boundaries

3.1 In Scope

- Web application with secure authentication and role-based access control.
- Multi-division operations for Davanagere, Shivamogga and Belagavi.

- Approximately 18 initial sections, with unlimited future extension.
- Department, section, category, brand, product, size, colour and dealer masters.
- Size/colour/variant quantity matrices.
- Purchase order creation, draft, submission, approval, issue, amendment and closure.
- Price, margin, discount, tax and additional-charge calculation.
- Partial and full goods receiving.
- Inventory transaction update on accepted receipt.
- Super Admin and Domain Admin dashboards.
- Reports, exports, comments, attachments and audit trail.
- Import/export facilities for master data and transactions.
- API-ready architecture for future integrations.

3.2 Phase 1 Boundaries

- Direct EDI/API integration with dealer ERP systems is deferred.
- POS billing is outside the procurement system.
- Bin-level warehouse put-away is deferred to a WMS layer.
- Advanced auto-replenishment depends on sales/POS data and is therefore future-phase functionality.

The uploaded source explicitly identifies direct dealer ERP integration, POS billing, and warehouse bin-level put-away as Phase 1 exclusions or later enhancements.

4. Business Context and Stakeholders

Stakeholder	Primary Responsibility	System Need
Super Admin / Head Office	Enterprise governance	All divisions, configuration, approvals, audit, consolidated reporting
Domain / Division Admin	Day-to-day branch procurement	Own division purchasing, section configuration within limits, dealers, approvals
Purchase Executive / User	PO data entry	Fast draft creation, product/variant quantity entry
Approver / Manager	Commercial governance	Review price, margin, discount and justification
Receiving / Inventory User	Goods receipt	Record received, damaged, rejected and pending quantities
Auditor	Control and verification	Read-only access to history and audit evidence
Finance / Accounts (future or read-only)	Financial reconciliation	PO value, supplier details, receipt and invoice linkage

5. Organizational Hierarchy and Configuration Model

The application shall use a centralized hierarchy in which transactional data is linked through a controlled organizational and product structure. The division selected at the beginning of the workflow determines the permitted data scope and reporting roll-up. The uploaded source specifies the hierarchy as organization → division → department → section → category → brand → product → variant.

System Organizational & Product Hierarchy

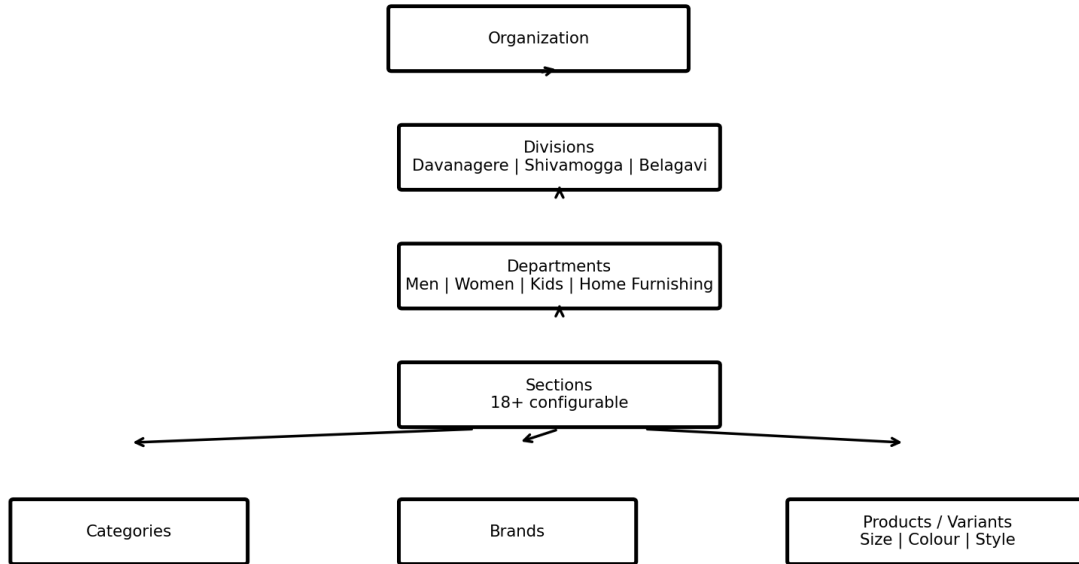


Figure 2. Division, department, section, brand and product hierarchy.

Level	Examples	Administrative Rules
Division	Davanagere, Shivamogga, Belagavi	Super Admin controls; users are assigned one or more scopes
Department	Men's, Women's, Kids, Home Furnishing	Admin configurable
Section	Men's Shirts, Sarees, Jewellery, etc.	Dynamic; add/edit/activate/deactivate/archive
Category/Subcategory	Shirts → Formal Shirts	Dynamic
Brand	Brand Number + Name + Serial	Unique identifiers and reusable master
Product	Product Serial + SKU + Name	Belongs to brand/category; supports variants
Variant	Size, Colour, Style, other attributes	Configured per product/section

6. Roles, Authentication and Authorization

6.1 Role Model

Role	Access Scope	Key Capabilities
Super Admin	All divisions and modules	Full configuration, user management, approvals, audit, reports, exports
Admin / Domain Admin	Assigned division(s)	Manage local purchasing, dealers, sections within allowed policy, division dashboard
Purchase Manager	Assigned division(s)/departments	Create/manage POs and review commercial details
Purchase Executive	Assigned division + section	Create draft POs; no final price/discount authority unless granted
Approver	Assigned approval queue	Approve/reject/send back based on configured rules
Receiving User	Assigned division	Create receipts and update received status
Viewer	Assigned reporting scope	Read-only dashboards/reports
Auditor	Audit scope	Read-only historical activity and audit evidence

6.2 Authentication Requirements

- Username/email + password authentication.
- Password reset and forced-reset capability.
- Account activation/deactivation.

- Session timeout and session invalidation.
- Optional MFA/OTP.
- Role resolution after authentication.
- Division-scope resolution before transaction access.
- Backend authorization on every protected API endpoint.

Security & Access Control Flow

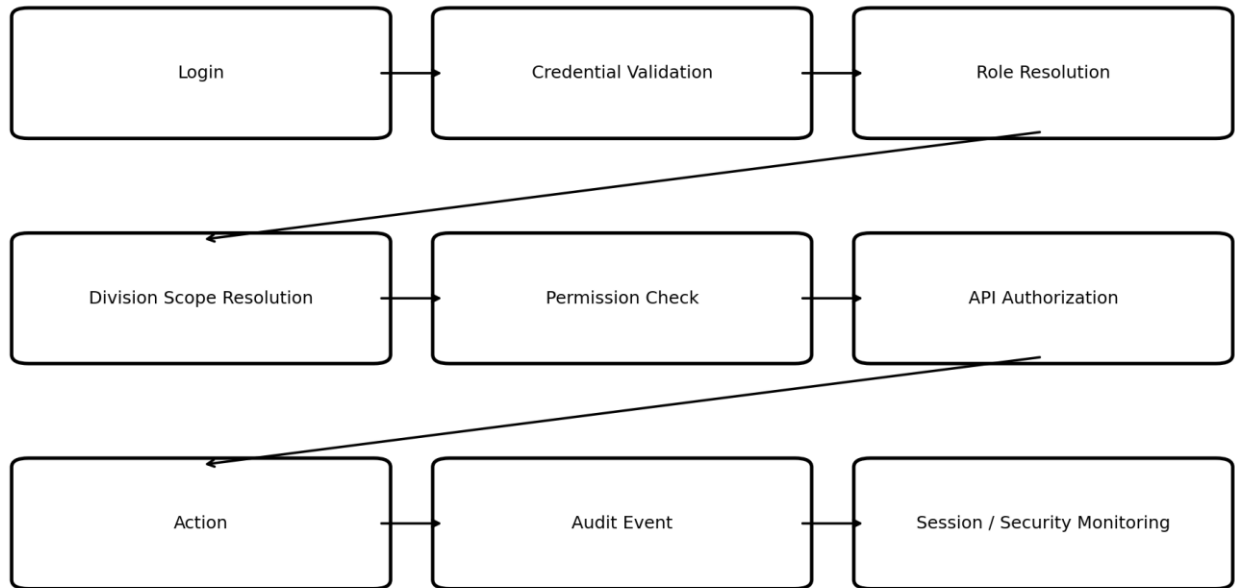


Figure 3. Security and authorization flow.

The source requirements call for role-based access control at both UI and API layers and secure password/session handling.

7. Master Data Management

Master data is the control layer of the application. The system shall avoid free-text duplication wherever controlled values exist. Administrators shall manage master data through dedicated screens, with audit logging and lifecycle states.

Master	Core Fields	Controls
Division	Code, name, location, contact, status	Unique code; scoped access
Department	Code, name, parent division or global flag	Active/inactive
Section	Name, parent group, sizing method, display order, attributes	Configurable; archive rather than destructive delete
Category	Code, name, parent section	Unique within parent
Brand	Brand number, name, serial, code, manufacturer, supplier	Duplicate prevention
Product	Serial, SKU, name, brand, section, attributes	Unique serial/SKU policy
Size	Method, label, order, numeric/code value	Section-specific sets
Colour	Name, code, family, swatch	Normalized values
Dealer/Supplier	Code, company, GSTIN/PAN, contacts, terms	Supplier lifecycle + history
Number Sequence	Prefix, year, division, running number	Concurrency-safe generation

8. Section Catalogue and the 18 Initial Sections + Furniture Extension

The source document defines 18 illustrative sections and requires the section structure to be data-driven. The system shall treat these as initial configuration rather than immutable code.

PURCHASE ORDER & MERCHANDISE PROCUREMENT MANAGEMENT SYSTEM

#	Section	Parent Group	Sizing Method
1	Men's Shirts	Men's Wear	S / M / L / XL / XXL / XXXL
2	Men's Trousers & Jeans	Men's Wear	Waist 28–44 + apparel sizes where applicable
3	Men's T-Shirts	Men's Wear	S / M / L / XL / XXL / XXXL
4	Men's Ethnic Wear (Kurta/Sherwani)	Men's Wear	S / M / L / XL / XXL / XXXL
5	Men's Innerwear	Men's Wear	S / M / L / XL / XXL
6	Women's Sarees	Women's Wear	Free Size / Custom
7	Women's Kurtis & Salwar Sets	Women's Wear	S / M / L / XL / XXL / XXXL
8	Women's Western Wear	Women's Wear	S / M / L / XL / XXL / XXXL
9	Women's Ethnic / Blouse Fabric	Women's Wear	Free Size / Stitched sizes
10	Women's Innerwear	Women's Wear	S / M / L / XL / XXL
11	Jewellery - Fashion / Artificial	Women's Wear → Jewellery	Free Size / Adjustable
12	Jewellery - Bangles & Sets	Women's Wear → Jewellery	Numeric bangle sizes
13	Kids Boys Wear	Kids	Age group 0–16 years
14	Kids Girls Wear	Kids	Age group 0–16 years
15	Kids Infant Wear	Kids	Age group 0–24 months
16	Footwear - Men	Men's Wear	UK size 6–12
17	Footwear - Women	Women's Wear	UK size 3–8
18	Accessories (Belts, Bags, Watches)	Men's / Women's Wear	Free Size / configurable
19	Furniture & Home Interiors	Home / Lifestyle	Configurable: Single / Set / Dimension / Custom

8.1 Home Furnishing Extension

Home Furnishing must be supported as a dedicated configurable section or department group. Typical items include bedsheets, blankets, pillows, pillow covers, curtains, towels, carpets, mats, cushions, table covers, sofa covers, rugs and kitchen textiles. Suggested attributes are material, dimensions, size, colour, pattern, unit, quantity, purchase price, discount, margin, tax, net value, dealer and remarks.

8.2A Furniture & Home Interiors Extension

Furniture is added as a first-class procurement section in this system. It is separate from Home Furnishing because furniture requires different attributes, dimensions, units, delivery considerations, assembly information and quantity handling than textile home-furnishing products.

Publicly available information about B.S. Channabasappa & Sons / BSC identifies a broad lifestyle and home-furnishing offering. The official company information specifically references dedicated ladies, kids, men's wear and home-furnishing spaces, while a public BSC Exclusive description also lists sarees, womenswear, kidswear, menswear, toys, accessories and home furnishings. The exact live furniture SKU assortment is not published as an exhaustive online catalogue; therefore, the furniture product taxonomy below is a system-ready procurement master structure and should be treated as configurable rather than a claim that every listed SKU is currently stocked at every outlet.

8.2B Furniture Product Sections and Detailed Product Catalogue

The furniture master shall be organized by functional area. Each category is configurable, and administrators may add, rename, deactivate or archive subcategories and products without a code deployment.

8.2C Furniture-Specific Purchase Order Fields

- Furniture Type / Category
- Brand Number, Brand Name and Brand Serial Number
- Product Serial Number, Product Code and SKU
- Material (wood / engineered board / metal / plastic / glass / mixed / custom)
- Finish / Surface / Upholstery
- Colour / Colour Family
- Dimensions: Length × Width × Height; depth where applicable

- Unit of Measure (piece, set, pair, unit, running foot where approved)
- Configuration / Variant / Model
- Assembly Required (Yes/No)
- Assembly Charges
- Delivery Lead Time
- Warranty / Guarantee Period
- Quantity
- Original Purchase Price
- Discount
- Margin
- Tax
- Additional Charges (freight, delivery, installation, assembly)
- Net / Final Value
- Dealer / Supplier
- Remarks and reference images/drawings

8.2D Furniture Quantity and Dimension Rules

- Furniture generally uses piece / set / pair quantities rather than clothing size matrices; however, configurable size or dimension variants must remain supported.
- When a furniture product has multiple dimensions or finishes, each unique variant must be entered as a separate line or variant combination.
- For made-to-order items, the purchase order must record the requested dimensions and attach a reference drawing/image when required.
- A product must not be received into inventory unless ordered quantity, accepted quantity and unit of measure are consistent.
- Delivery, installation and assembly charges must be recorded separately from the base furniture purchase price when they are supplier-billed.

8.2E Furniture Procurement Flow Diagram

8.2F BSC Exclusive Procurement Coverage Matrix

8.2 Add / Edit / Activate / Archive / Delete Behaviour

- Add Section: requires name, parent group, sizing method, display order and optional icon/image.
- Edit Section: changes configuration for future transactions while preserving historical transaction snapshots.
- Deactivate Section: removes it from new PO selection but preserves records.
- Archive Section: historical-safe retirement status.
- Delete Section: hard delete only if there are no dependencies or transactions; otherwise convert to archive.
- Any lifecycle change must be audit logged with user, timestamp, division, previous state and new state.

9. Size, Colour and Variant Management

9.1 Size Methods

Sizing Method	Examples	Typical Use
Standard Apparel	XS, S, M, L, XL, XXL, XXXL	Shirts, T-shirts, Kurtis, ethnic wear
Numeric Waist	28, 30, 32, 34, 36, 38, 40, 42, 44	Trousers and jeans
Free Size	Single Free Size quantity	Sarees, many jewellery items
Age Group	0-2, 2-4, 4-6, 6-8, 8-10, 10-12, 12-14, 14-16 years	Kids
Footwear UK	3-12	Footwear
Custom	Admin-defined label/value list	Future sections

9.2 Colour Master

- Maintain normalized colour names and codes.
- Support colour families and optional swatches.
- Allow Other → manual colour entry.
- Prevent spelling-based duplicates during creation.

9.3 Variant Matrix

For apparel the preferred PO entry mode is a colour-by-size matrix, for example Black → S/M/L/XL/XXL/XXXL quantities. The system must calculate total units automatically and validate that the sum of size quantities matches the stored total. The uploaded source explicitly requires size-wise quantity capture and a running total in the PO matrix.

10. Product and Brand Management

10.1 Brand Master

- Brand Number: unique business identifier.
- Brand Name: human-readable display name.
- Brand Serial Number: independent internal reference.
- Brand Code: short system code where required.
- Manufacturer and linked suppliers.
- Status and lifecycle dates.

10.2 Product Master

- Product Serial Number: system-generated unique reference.
- SKU and optional barcode.
- Product code, name and description.
- Brand references.
- Department, section, category and subcategory.
- Material/fabric, pattern, style, season, gender and age group where applicable.
- Tax category/HSN-SAC where applicable.
- Images and attachments.

10.3 Other Option / Progressive Catalogue

Brand, Product, Colour and Dealer selection must include an Other option where permitted. The Other workflow opens an inline form rather than sending the user away from the PO. A manual entry should be validated, flagged as custom data, and optionally promoted into the corresponding master after submission/approval. This progressive-catalogue approach is explicitly described in the supplied requirements.

11. Dealer / Supplier Management

11.1 Dealer Master Fields

- Dealer ID and Dealer Code.
- Dealer/company name and contact person.
- Mobile, email and address.
- GSTIN and PAN where applicable.

- Division association (one or more divisions).
- Payment terms: Advance, 30 Days, 60 Days, Against Delivery or custom.
- Preferred brands/categories.
- Status, notes and internal rating.
- Optional bank details subject to permissions and future payment integration.

11.2 Dealer Ledger

- Total POs raised.
- Total ordered value.
- Total received value.
- Outstanding balance (when finance linkage is implemented).
- PO history and statuses.
- Delivery performance and quality metrics in future phases.

12. Purchase Order Functional Requirements

12.1 PO Header

Field Group	Required Data
Identification	PO Number, PO Date, Version/Revision, Status
Organization	Division, Department, Section
Ownership	Created By, Requesting User, Buyer
Supplier	Dealer Code, Dealer Name, Contact, Address
Commercial	Payment Terms, Delivery Terms, Currency, Tax Scheme
Schedule	Expected Delivery Date
Control	Approval Status, Created/Updated Timestamps
Supporting Data	Remarks, attachments, internal comments

12.2 Guided PO Creation

1. Authenticate user and resolve authorized division(s).
2. Select Division: Davanagere / Shivamogga / Belagavi or another division permitted by policy.
3. Select Department and active Section.
4. Select or add Dealer/Supplier.
5. Select Brand; use Other when necessary.
6. Select Product/Style; use Other when necessary.
7. Select Colour; use Other when necessary.
8. Enter size-wise or attribute-wise quantity.
9. Enter original purchase price and margin.
10. Apply discount; add tax/charges where configured.
11. Review system-calculated totals and commercial validation.
12. Save Draft or Submit for approval.

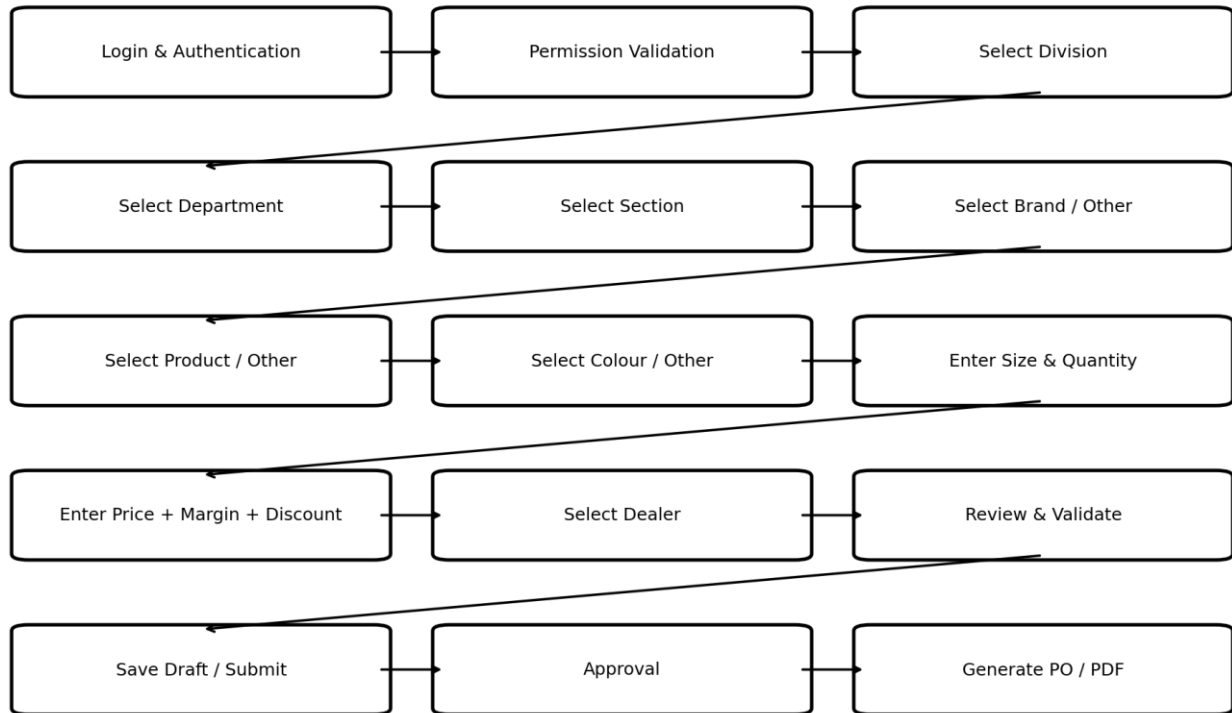
User / Purchase Entry Flow

Figure 4. Guided purchase entry workflow.

12.3 Product Entry Grid

Product	Brand	Colour	S	M	L	XL	XXL	XXXL	Total
Formal Shirt	Brand A	Black	5	10	12	10	6	2	45
Formal Shirt	Brand A	White	3	7	9	8	4	1	32

The grid must support keyboard navigation, copy/duplicate row, add/remove row, live total calculation and validation. Different sections may use different matrix columns based on their configured sizing method.

13. Pricing, Margin, Discount, Tax and Charges

The uploaded source defines the commercial formula as: Net Value = Purchase Price + (Purchase Price × Margin %); Discount Amount = Net Value × Discount % (or a flat amount); Final Value = Net Value - Discount Amount; Line Total = Final Value × Total Quantity.

Calculation	Formula
Gross Value	Purchase Price × Total Quantity
Margin Amount	Purchase Price × Margin %
Net Value / Selling Value per Unit	Purchase Price + Margin Amount
Discount Amount	Net Value × Discount % OR Flat Discount
Final Value per Unit	Net Value – Discount Amount
Line Total	Final Value per Unit × Total Quantity
Subtotal	Sum of line totals before order-level charges
Tax	Configured tax rule on applicable taxable base
Additional Charges	Freight + packing + transport + other configured charges
Grand Total	Subtotal – order-level discount + tax + charges

13.1 Worked Example

Field	Value
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PURCHASE ORDER & MERCHANDISE PROCUREMENT MANAGEMENT SYSTEM

Brand / Product	Levi's - Men's Formal Shirt
Colour	Navy Blue
Size Qty	S10 / M20 / L20 / XL15 / XXL10 / XXXL5 = 80
Purchase Price / Unit	₹650.00
Margin	40% = ₹260.00
Net Value / Unit	₹910.00
Discount	10% = ₹91.00
Final Value / Unit	₹819.00
Line Total	₹65,520.00

All calculation fields must be stored with appropriate monetary precision and recalculated server-side. The UI may preview the result, but the backend must remain authoritative.

14. Approval and Governance Workflow

The system must support configurable approval levels. Approval rules may depend on PO value, division, department, section, dealer, category, discount or role. Example thresholds in the source document are ₹0–₹25,000 → Admin Approval; ₹25,001–₹1,00,000 → Division Manager Approval; above ₹1,00,000 → Super Admin Approval. These are examples and must be configurable.

Configurable Approval Flow

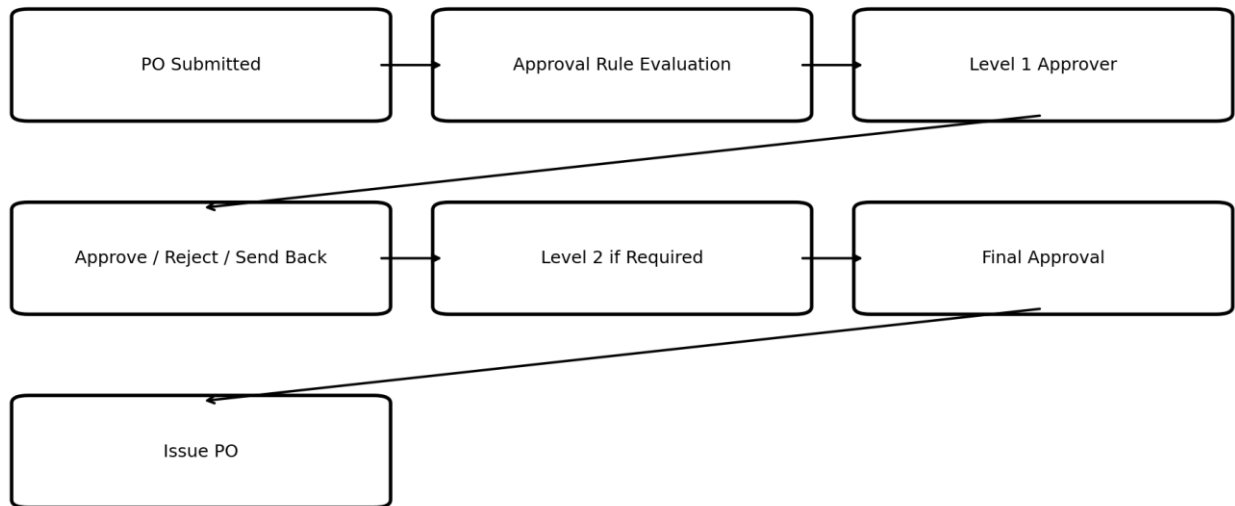


Figure 5. Approval engine and escalation flow.

14.1 Approval Actions

- Approve.
- Reject (reason mandatory).
- Send Back / Request Changes (comment mandatory).
- Put on Hold.
- Escalate when configured.

14.2 Approved PO Control

- Approved POs become immutable for ordinary editing.
- Any change requires an amendment/version event.
- Commercial changes may automatically trigger re-approval.
- Each version keeps the previous version accessible.

15. Receiving and Inventory Integration

Receiving must operate against the issued PO and preserve ordered, received, rejected and pending quantities. Partial receipt is a first-class state rather than an exception.

Quantity State	Meaning
Ordered	Original PO quantity
Accepted Received	Quantity accepted into inventory
Damaged	Physically received but not accepted into stock
Rejected	Not accepted against PO
Pending	Ordered minus accepted/rejected/closed handling as configured

Receiving & Inventory Flow

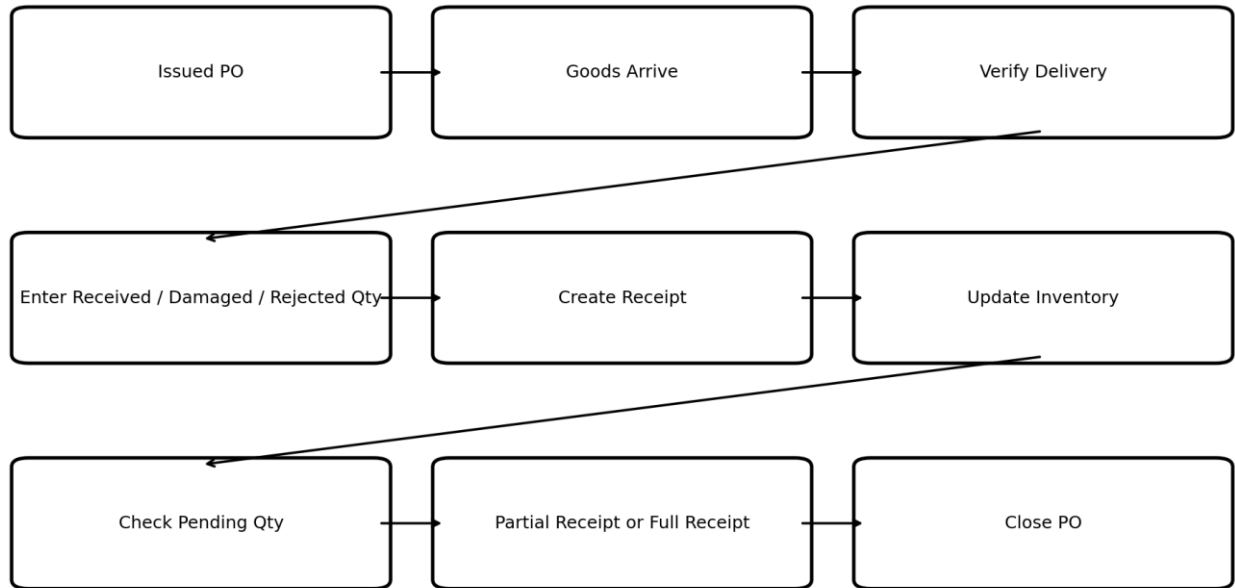


Figure 6. Goods receiving and inventory update flow.

15.1 Receipt Header

- Receipt Number.
- PO Number.
- Supplier.
- Delivery date.
- Invoice number/date where available.
- Received by.
- Division.
- Remarks and attachments.

15.2 Receipt Line

- Product/variant reference.
- Ordered quantity.
- Received quantity.
- Damaged quantity.
- Rejected quantity.

- Accepted quantity.
- Remarks.

16. Dashboards, Reporting and Analytics

16.1 Super Admin Dashboard

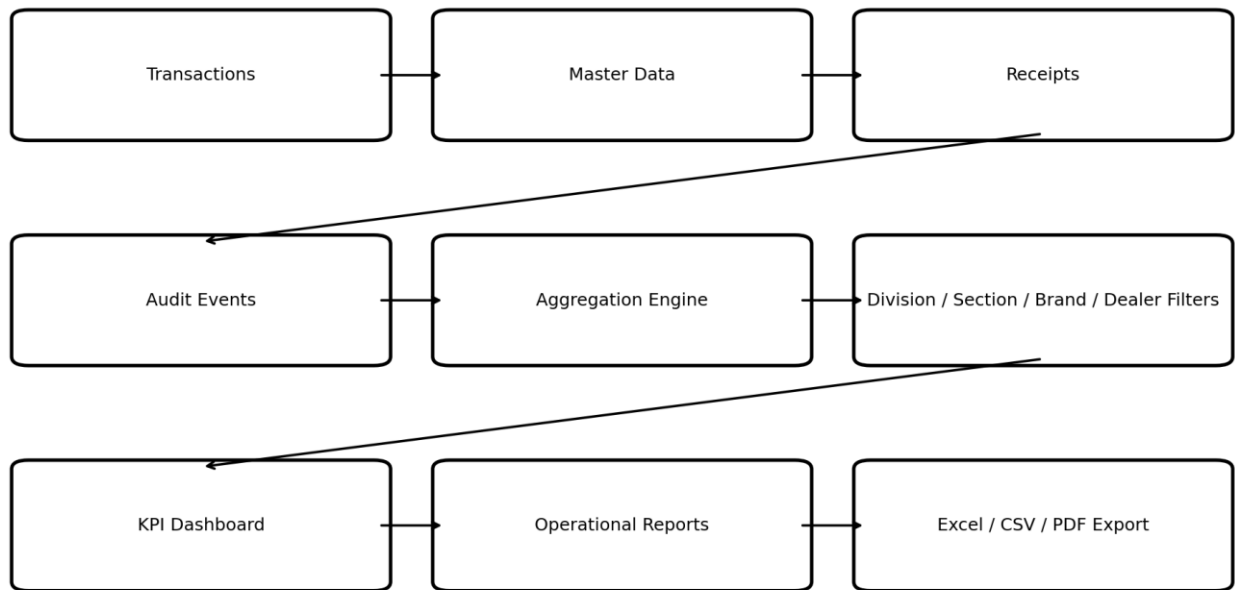
- Total POs, pending approvals, approved, issued, partial and completed.
- Total quantity and total purchase value.
- Total discount and margin measures.
- Section, brand and dealer breakdown.
- Davanagere vs Shivamogga vs Belagavi comparison.
- Pending approval queue.
- Full enterprise audit activity.
- Drill-down into any division.

16.2 Domain Admin Dashboard

- Own-division POs and values.
- Pending approvals.
- Active sections and dealers.
- Section/brand spend.
- Size-curve analysis.
- Own-division audit history.

16.3 Reports

Report	Primary Dimensions
PO Register	PO, date, division, dealer, status, value
Purchase Summary	Division, department, section, period
Brand Report	Brand, product, quantity, value
Dealer Report	Dealer, POs, ordered/received value
Size Report	Product/brand/size quantity
Colour Report	Product/colour quantity
Discount Report	PO, dealer, discount type/value
Margin Report	Product, section, division, margin
Approval Ageing	PO, approver, submitted date, pending duration
Receiving Report	Ordered vs received vs pending
Audit Report	User, action, timestamp, before/after
Amendment Report	PO version history

Analytics & Reporting Flow*Figure 7. Transaction-to-reporting flow.*

17. Audit Trail, History and Document Management

Every critical action must be logged in an immutable audit stream. The uploaded requirements explicitly require all edits, section changes, master additions and commercial edits to be visible to the appropriate administrators.

Audit Field	Description
Event ID	Unique audit event identifier
Timestamp	Exact date/time
User + Role	Actor and resolved role
Division + Section	Business scope
Action Type	Create/Edit/Delete/Approve/Receive/etc.
Record Type + ID	Target entity
Before Value	Previous value where changed
After Value	New value where changed
Reason / Comment	Justification for sensitive action
Request/Correlation ID	Technical trace for API/server diagnostics

17.1 Attachments and Comments

- Quotation attachments.
- Supplier documents.
- Invoice or delivery proof.
- Product/reference images.
- Approval notes.
- Internal comments with user and timestamp.

17.2 PO Timeline

Each PO should show a chronological timeline such as Created → Submitted → Reviewed → Approved → Issued → Receipt 1 → Receipt 2 → Closed, with actor, timestamp and action details.

18. Notifications and Communication

- PO submitted for approval.
- Approval pending.
- PO approved.
- PO rejected/send back.
- PO issued.
- Delivery due/overdue.
- Partial receipt.
- Full receipt.
- High-value or high-discount alert.
- Master-data creation awaiting authorization where configured.

Phase 1 may use in-app and email notifications. Future phases can add SMS/WhatsApp and vendor portal communications.

19. UI/UX and Screen Specifications

19.1 Main Navigation

- Dashboard
- Purchasing → Purchase Orders / Approvals / Receiving
- Masters → Divisions / Departments / Sections / Brands / Products / Sizes / Colours / Dealers
- Inventory
- Reports
- Administration → Users / Roles / Audit Logs / Settings

19.2 Key Screens

Screen	Primary Components	Key Actions
Login	Credentials, reset, MFA option	Sign in, reset password
Division Selection	Authorized division cards/list	Select division
Dashboard	KPIs, charts, activity	Drill down, filter
Section Management	Tree/list, Add Section	Create/edit/archive/configure
Brand/Product Masters	Searchable tables + forms	CRUD, import, export
PO Create	Guided stepper + matrix	Add lines, calculate, save, submit
PO Details	Tabs: overview/products/approval/receiving/history	View, approve, amend, receive
Approval Queue	Pending cards/table	Approve/reject/send back
Receiving	PO selector + receipt grid	Record receipt
Reports	Filters, tables, charts	Export
Users & Roles	Users, scopes, permissions	Create/assign/deactivate
Audit Logs	Immutable event table	Filter, export, drill into record

19.3 UX Standards

- Desktop-first, tablet-friendly, responsive mobile fallback.
- Searchable dropdowns and type-ahead.
- Fast matrix entry for quantities.
- Keyboard navigation for high-volume entry.
- Sticky totals and validation.
- Explicit confirmation for destructive actions.
- Clear status chips and workflow indicators.
- No inaccessible dead buttons or fake dashboard metrics.

20. Data Model and Database Requirements

The database shall be normalized enough to prevent duplication while retaining transaction snapshots so that historical PO documents do not change when master records are updated later.

20.1 Core Entities

Entity	Important Relationships
users	roles, user_divisions, departments
roles	permissions, users
divisions	departments, sections, users, POs
departments	sections, products
sections	categories, sizing methods, attributes, POs
categories	products
brands	products, suppliers
products	brand, section, variants
product_variants	product, size, colour
sizes	size methods, variants
colours	variants
suppliers	brands, divisions, POs
purchase_orders	division, supplier, user, approvals, receipts
purchase_order_items	PO, product/variant, quantity, pricing
approvals	PO, approver, approval level
receipts	PO, receipt lines
inventory_transactions	product variant, receipt, division
notifications	user, entity/event
attachments	entity reference
audit_logs	actor, target entity, before/after

20.2 Suggested Table Set

- users, roles, permissions, user_roles, user_divisions
- divisions, departments, sections, categories, section_attributes
- brands, products, product_variants, sizes, size_methods, colours
- suppliers, supplier_brands
- purchase_orders, purchase_order_items, purchase_order_quantities
- purchase_order_discounts, purchase_order_taxes, purchase_order_charges
- approval_rules, approval_levels, approval_instances
- receipts, receipt_items, inventory_transactions
- notifications, attachments, comments, audit_logs
- number_sequences, settings, import_jobs, export_jobs

20.3 Historical Snapshot Principle

At approval/issue time, the PO should preserve the relevant display values (brand name, product name, colour label, size label, dealer information and commercial values) in transactional snapshots. Master data may evolve later, but historical PO documents must remain reproducible.

21. API and Integration Requirements

Domain	Representative Endpoints
Authentication	/api/auth/login, /api/auth/refresh, /api/auth/logout, /api/auth/reset
Users/Roles	/api/users, /api/roles, /api/permissions
Organization	/api/divisions, /api/departments, /api/sections
Catalogue	/api/brands, /api/products, /api/categories, /api/sizes, /api/colours
Suppliers	/api/suppliers, /api/suppliers/:id/history
Purchasing	/api/purchase-orders, /api/purchase-orders/:id, /api/purchase-orders/:id/submit
Approvals	/api/approvals, /api/approvals/:id/approve, /reject, /send-back
Receiving	/api/receipts, /api/receipts/:id
Inventory	/api/inventory/transactions, /api/inventory/balances

Reports	/api/reports/*
Audit	/api/audit-logs
Notifications	/api/notifications

- Use pagination, filtering and sorting on list APIs.
- Return consistent error structures.
- Authorize every endpoint server-side.
- Use idempotency for sensitive create operations where appropriate.
- Protect number sequence generation from race conditions.
- Provide API versioning for future external integrations.

22. Security Requirements

- Passwords must be securely hashed; never stored in plaintext.
- HTTPS required in production.
- Input validation and output encoding.
- Protection against SQL injection, XSS, CSRF where applicable and insecure direct object references.
- Rate limiting on authentication and sensitive endpoints.
- Least-privilege role and division access.
- Audit logging of privileged operations.
- Re-authentication/step-up verification for high-risk actions where configured.
- File-upload validation, size/type limits and malware scanning strategy.
- Secrets stored outside source code in secure environment configuration.
- Session expiration and revocation.
- No client-supplied calculation may be trusted without backend recomputation.

23. Non-Functional Requirements

Area	Requirement
Performance	Dashboard and common list operations should be responsive; source target is consolidated trailing-12-month dashboard load within about 3 seconds.
Availability	Production deployment should use monitoring, health checks and backup/recovery procedures.
Scalability	Support growth in divisions, products, brands, dealers and PO history without redesign.
Usability	Guided PO flow, type-ahead searches, matrix entry and clear validation.
Data Integrity	Foreign keys, unique constraints, transactional operations and server-side calculation.
Auditability	Immutable audit event recording for critical operations.
Accessibility	Clear labels, keyboard navigation, readable contrast, usable error messages.
Maintainability	Modular domain services, reusable components, configuration-driven rules.
Observability	Application logs, API error logs, background job logs and security events.

24. Business Rules and Validation Rules

- RB-001: User may access only divisions granted by authorization policy.
- RB-002: Only active divisions, departments and sections may be selected for new POs.
- RB-003: Historical POs must remain readable after section or master-data retirement.
- RB-004: Product serial number and SKU uniqueness must be enforced according to configured scope.
- RB-005: Brand number and brand serial number must not be conflated.
- RB-006: Total quantity equals the sum of configured variant quantities.
- RB-007: Quantity cannot be negative; zero quantity lines are invalid unless explicitly used for template scenarios.
- RB-008: Purchase price cannot be negative.
- RB-009: Discount cannot exceed allowed policy without exception approval.
- RB-010: Margin cannot exceed configured policy without permission/approval.
- RB-011: Approved POs are not directly editable.

24. RB-012: Rejection and send-back actions require reasons/comments.
25. RB-013: Receipt quantities must reconcile with PO ordered quantities.
26. RB-014: A section with transaction history is archived/soft-deleted rather than physically erased.
27. RB-015: All critical state transitions are audit logged.
28. RB-016: Other/manual values must be visibly flagged as custom until promoted to controlled master data.
29. RB-017: Financial totals must be recalculated server-side before persistence/approval.
30. RB-018: Division-scoped reports must not leak cross-division data.

25. Exception and Edge-Case Handling

Scenario	Expected Behaviour
Duplicate brand found	Warn, show matching records, prevent duplicate where unique key matches
Duplicate product serial/SKU	Hard validation error unless controlled exception policy exists
Section has history and user clicks delete	Offer archive/deactivate instead of hard delete
User loses session during PO	Autosave draft where feasible; require re-authentication on resume
Approval rule unavailable	Do not issue PO; route to configured fallback or Super Admin exception queue
Supplier becomes inactive after draft	Existing draft remains readable, but new submission requires active supplier or authorized override
Partial delivery	Create receipt and keep PO Partially Received
Over-receipt	Block or require approved over-receipt exception
Price changes after draft	Recalculate and record change; approval rule may restart
Network interruption on submit	Use idempotent submission to avoid duplicate PO creation
Import file contains errors	Reject invalid rows, provide row-level error report, do not silently insert

26. Testing and Acceptance Criteria

26.1 Test Categories

- Unit tests for calculations, validation and business rules.
- Integration tests for PO → approval → receipt → inventory.
- API contract tests for protected endpoints.
- UI workflow tests for core screens.
- Security tests for role and division bypass attempts.
- Import/export tests.
- PDF generation tests.
- Performance tests for dashboard and large catalogue searches.
- Regression tests for section/configuration changes.

26.2 Key Acceptance Scenarios

31. Create a user and assign a role and division.
32. Login and verify only authorized division(s) are visible.
33. Add/edit/archive a section without code deployment.
34. Configure S-XXXL sizes and confirm the matrix updates automatically.
35. Create a brand with separate brand number, name and brand serial number.
36. Create a product with product serial number and variant definitions.
37. Use Other for brand/product/colour/dealer and verify controlled promotion behaviour.
38. Create a multi-line PO with colour × size quantities.
39. Apply margin and discount and verify server-side calculations.
40. Route a high-value PO to the correct approval level.
41. Reject/send back a PO and verify reason capture.
42. Approve and issue the PO; generate PDF.
43. Partially receive a PO and verify remaining quantity.

44. Fully receive the PO and verify close/receipt state.
45. Confirm dashboards and reports reflect the transaction.
46. Confirm audit log contains user, time, action and before/after values.

27. Implementation Phases and Delivery Plan

Phase	Scope	Exit Criteria
Phase 0 - Discovery	Finalize master data policy, approval policy, identifiers, UI prototypes	Signed-off business rules and UX
Phase 1 - Core Foundation	Auth, roles, divisions, sections, masters, PO engine	Core PO lifecycle works end-to-end
Phase 2 - Governance	Approvals, audit, amendments, notifications, PDF	Governed PO process production-ready
Phase 3 - Receiving	Receipts, partial receiving, inventory transactions	Receipt lifecycle reconciles to PO
Phase 4 - Analytics	Dashboards, reports, exports, drill-down	Management reporting validated
Phase 5 - Hardening	Security, performance, backup, observability, UAT	Production readiness sign-off
Phase 6 - Optional Extensions	Vendor portal, barcode, auto-replenishment, accounting/ERP integrations	Prioritized enhancements delivered separately

Administration & Configuration Flow

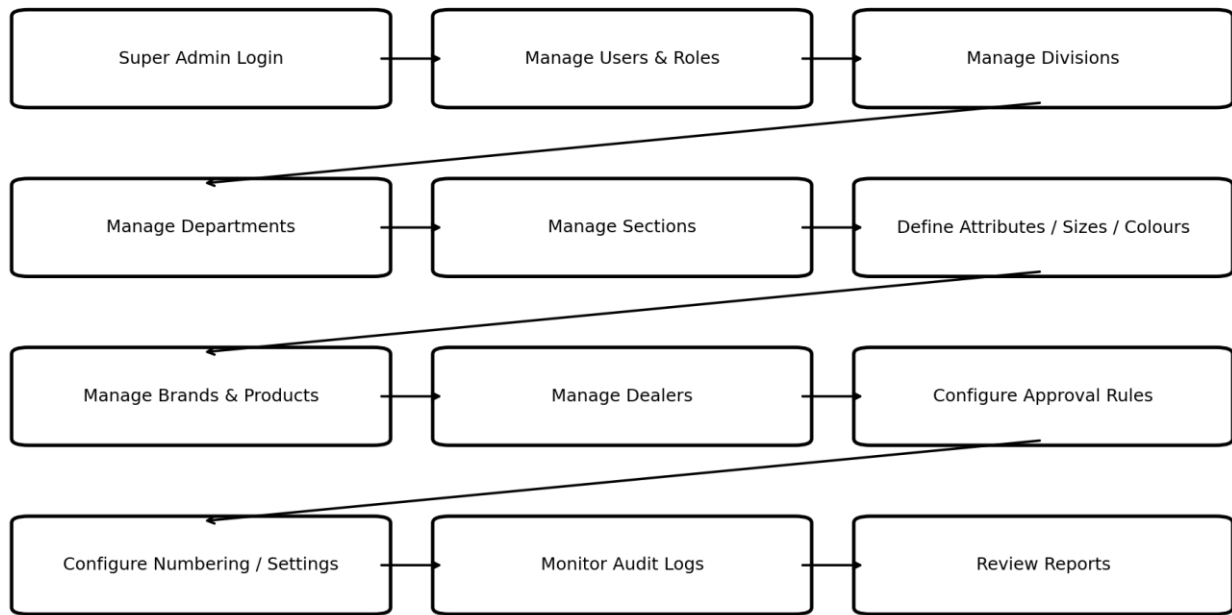


Figure 8. Administration and configuration flow.

28. Deployment, Backup and Operations

- Separate Development, Staging and Production environments.
- HTTPS and secure DNS in production.
- Environment variables for secrets/configuration.
- Database backup schedule with tested restore procedure.
- At least 30-day backup retention as the source baseline.
- Monitoring for application health, database health, API errors and background jobs.
- Controlled schema migrations.
- Centralized logs and alerting.
- Disaster recovery runbook and periodic restore test.

29. Future Enhancements and Market Benchmark Features

The uploaded requirements benchmark the solution against apparel and retail procurement systems and identify future-facing features such as size-curve matrices, digital vendor confirmation, barcode/SKU generation, GST-compliant invoicing, vendor portals, auto-replenishment, real-time PO status and multi-outlet reporting. The source recommends prioritizing core workflow first and scheduling vendor portal, barcode generation and auto-replenishment after stabilization.

Furniture extension note: the procurement architecture is intentionally generic enough to support both stocked furniture and made-to-order/custom furniture, including dimension capture, reference drawings, delivery/installation charges, warranty details and variant-based pricing.

Future Feature	Business Value	Recommended Phase
Vendor Portal	Dealer can view/confirm/reject PO and proposed changes	Phase 2+
Barcode/SKU Generation	Automated style-colour-size identification	Phase 2+
Supplier Quotation Comparison	Compare supplier offers before PO	Phase 2+
Auto Replenishment	Suggested reorder quantities from stock/sales	Phase 3+
Price History	Historical purchase price trend by product/dealer	Phase 2+
Supplier Ranking	Delivery, quality and price scoring	Phase 2+
Three-Way Matching	PO vs receipt vs invoice	Phase 3+
Accounting / ERP Integration	Financial posting and reconciliation	Phase 3+
Native Mobile App	On-the-go approvals and PO review	Later
Advanced Forecasting	Seasonality and demand prediction	Later

30. Glossary

Term	Definition
Division	Operating branch such as Davanagere, Shivamogga or Belagavi.
Department	High-level merchandise group such as Men's, Women's or Kids.
Section	Configurable purchasing category such as Men's Shirts or Sarees.
Variant	Specific combination of product attributes such as colour and size.
Brand Number	Business identifier for a brand.
Brand Serial Number	Internal reference distinct from the brand number and name.
Product Serial Number	Unique identifier assigned to a product.
PO	Purchase Order issued to a supplier/dealer.
Net Value	Purchase price plus configured margin before discount under the supplied formula.
Final Value	Net value after discount.
Domain Admin	Administrator responsible for a defined division scope.
Super Admin	Head-office role with enterprise-wide access.
Receipt	Transaction recording goods received against a PO.
Audit Log	Immutable record of important system activity and changes.

31. Final End-to-End Flow

Purchase Order Lifecycle

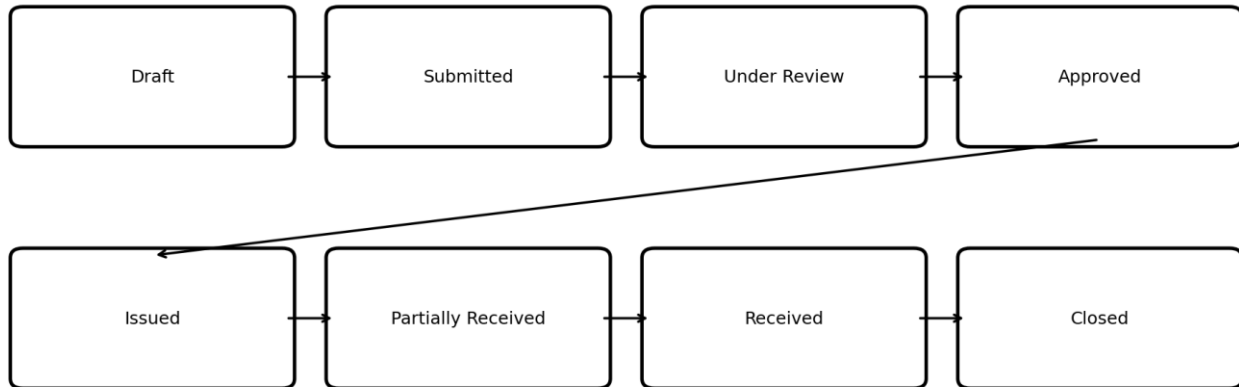


Figure 9. Purchase order lifecycle.

The final operating flow is: Login → Role/Permission Validation → Division → Department → Section → Brand/Product/Colour → Size/Quantity → Purchase Price → Margin → Discount → Dealer → Review → Draft/Submit → Approval → Issue PO → PDF/Print → Goods Receipt → Inventory Update → Dashboard/Reports → Audit History. This is consistent with the core workflow described in the supplied project requirements.

Appendix A - Screen Inventory

#	Screen
1	Login
2	Forgot Password / Reset Password
3	Division Selection
4	Super Admin Dashboard
5	Domain Admin Dashboard
6	Purchase Dashboard
7	PO List
8	PO Create
9	PO Edit Draft
10	PO Details
11	Approval Queue
12	Receipt List
13	Receipt Create
14	Section Management
15	Department Management
16	Category Management
17	Brand Management
18	Product Management
19	Size Management
20	Colour Management
21	Dealer/Supplier Management
22	User Management
23	Role & Permissions
24	Notifications
25	Reports
26	Audit Logs
27	Settings
28	Import Wizard
29	Export Center

Appendix B - Suggested API Inventory

#	Endpoint
1	POST /api/auth/login
2	POST /api/auth/logout
3	POST /api/auth/refresh
4	GET /api/me
5	GET/POST/PATCH /api/users
6	GET/POST/PATCH /api/divisions
7	GET/POST/PATCH /api/departments
8	GET/POST/PATCH /api/sections
9	GET/POST/PATCH /api/categories
10	GET/POST/PATCH /api/brands
11	GET/POST/PATCH /api/products
12	GET/POST/PATCH /api/sizes
13	GET/POST/PATCH /api/colours
14	GET/POST/PATCH /api/suppliers
15	GET/POST /api/purchase-orders
16	GET/PATCH /api/purchase-orders/{id}
17	POST /api/purchase-orders/{id}/submit
18	POST /api/purchase-orders/{id}/approve
19	POST /api/purchase-orders/{id}/reject
20	POST /api/purchase-orders/{id}/send-back
21	POST /api/purchase-orders/{id}/amend
22	GET/POST /api/receipts
23	GET /api/inventory/*
24	GET /api/reports/*
25	GET /api/audit-logs
26	GET /api/notifications

Appendix C - Sample PO Data Structure

```

PurchaseOrder
├── poNumber
├── divisionId
├── departmentId
├── sectionId
├── supplierId
├── status
├── approvalStatus
├── commercialSummary
│   ├── subtotal
│   ├── discount
│   ├── tax
│   ├── charges
│   └── grandTotal
└── lines[]
    ├── brandId / brandSnapshot
    ├── productId / productSnapshot
    ├── colourId / colourSnapshot
    ├── variantQuantities[]
    │   ├── sizeId / sizeLabel
    │   └── quantity
    ├── purchasePrice
    ├── marginPercent
    ├── marginAmount
    ├── netValue
    ├── discountType
    ├── discountAmount
    └── lineTotal
  
```

Appendix D - Implementation Definition of Done

- All required modules are implemented and connected to persistent data.
- No critical workflow uses placeholder/mock data in production mode.
- Frontend and backend permission checks are aligned.

- Server-side financial calculations match approved formulas.
- Approved PO versioning is implemented.
- Receipt quantities reconcile correctly.
- Audit entries are generated and queryable.
- PDF generation is stable.
- Import/export validations are complete.
- Critical test suite passes.
- Security review and UAT are completed.
- Backup and restore procedures are documented and tested.

Final Project Statement

This document is intended to act as the authoritative build blueprint for the Purchase Order and Merchandise Procurement Management System, including BSC Exclusive furniture and lifestyle procurement. The application should be implemented as a configurable enterprise platform rather than as a set of hard-coded category pages. The core architecture must support the three current divisions, approximately 18 initial sections, dynamic future sections, product variants, size/colour matrices, commercial calculations, dealer management, approval governance, receiving, inventory integration, dashboards, reporting, auditability and future ERP/vendor integrations.

Publicly Identified BSC Category	Evidence / Meaning	System Treatment
Men's Wear	Official company site identifies men's wear / Men's Xpress formats.	Department > Men's; full brand/product/variant master.
Women's / Ladies Wear	Official company site identifies dedicated ladies spaces.	Department > Women's; apparel + saree + accessories.
Kids Wear	Official company site identifies dedicated kids spaces.	Department > Kids; age/size matrices.
Sarees	Public BSC listings identify sarees and multiple saree types.	Women's > Sarees; fabric/pattern/colour/length fields.
Jewellery / Accessories	System requirement includes jewellery; public listings also identify accessories.	Women's / Accessories with configurable attributes.
Toys	A public BSC Exclusive description identifies toys.	Kids > Toys; age group, material, quantity, safety/document fields.
Home Furnishings	Official company information and local public descriptions identify home furnishings.	Home > Home Furnishing; textile and decor product attributes.

Furniture Section	Product Categories	Representative Products	Key Procurement Attributes
Living Room	Sofas & Seating	3-seater sofa, 2-seater sofa, sectional/L-sofa, recliner, accent chair, lounge chair, ottoman, sofa set	Material, upholstery, colour, seating capacity, dimensions, finish, warranty
Tables	Center / Coffee / Side / Console	center table, coffee table, side table, nesting table, console table, end table	Length, width, height, material, finish, shape, top type
TV & Entertainment	TV Units & Media Storage	TV unit, media console, wall-mounted unit, entertainment cabinet	TV-size compatibility, dimensions, material, cable management, finish
Bedroom	Beds & Bed Frames	single bed, queen bed, king bed, storage bed, hydraulic bed, bunk bed	Mattress size, outer dimensions, material, storage type, finish
Bedroom Storage	Wardrobes & Chests	2-door wardrobe, 3-door wardrobe, sliding wardrobe, chest of drawers, dresser	Dimensions, doors, shelves, drawers, mirror, material, finish
Bedside & Dressing	Nightstands / Dressers	bedside table, nightstand, dressing table, dresser, mirror cabinet	Dimensions, drawer count, mirror type, material, finish
Dining	Dining Sets	4-seater, 6-seater, 8-seater dining table/set, benches	Seating capacity, table dimensions, tabletop material, chair material, finish
Dining Seating	Dining Chairs & Stools	dining chair, bar stool, counter stool, bench	Height, upholstery, material, load capacity, finish
Study / Office	Work & Study Furniture	study table, computer table, office desk, executive desk, workstation	Desk dimensions, storage, cable management, material, finish
Office Seating	Office Chairs	task chair, executive chair, visitor chair, meeting chair	Chair type, upholstery, height adjustment, arms, wheel/base type
Storage	Cabinets & Shelving	bookshelf, display unit, crockery cabinet, utility cabinet, filing cabinet	Dimensions, shelves, doors, lock, load capacity, material
Entryway	Entrance Furniture	shoe rack, console, bench, key cabinet, storage bench	Capacity, dimensions, doors/drawers, material, finish
Kids Furniture	Children's Furniture	kids bed, study table, study chair, toy storage, kids wardrobe, bunk bed	Age group, dimensions, safety features, material, finish
Outdoor / Patio	Outdoor Furniture	patio chair, outdoor table, swing, bench, garden set	Material, weather resistance, dimensions, finish, assembly
Modular / Custom	Configured or Made-to-Order	modular wardrobe, TV wall unit, modular storage, customized table/cabinet	Custom dimensions, drawing/reference image, material, laminate/finish, lead time

Furniture Section	Product Categories	Representative Products	Key Procurement Attributes
Furniture Accessories	Supporting Items	cushions, chair pads, furniture covers, hardware kits, legs, handles	Size, material, compatibility, quantity, finish

Furniture Procurement Extension Flow

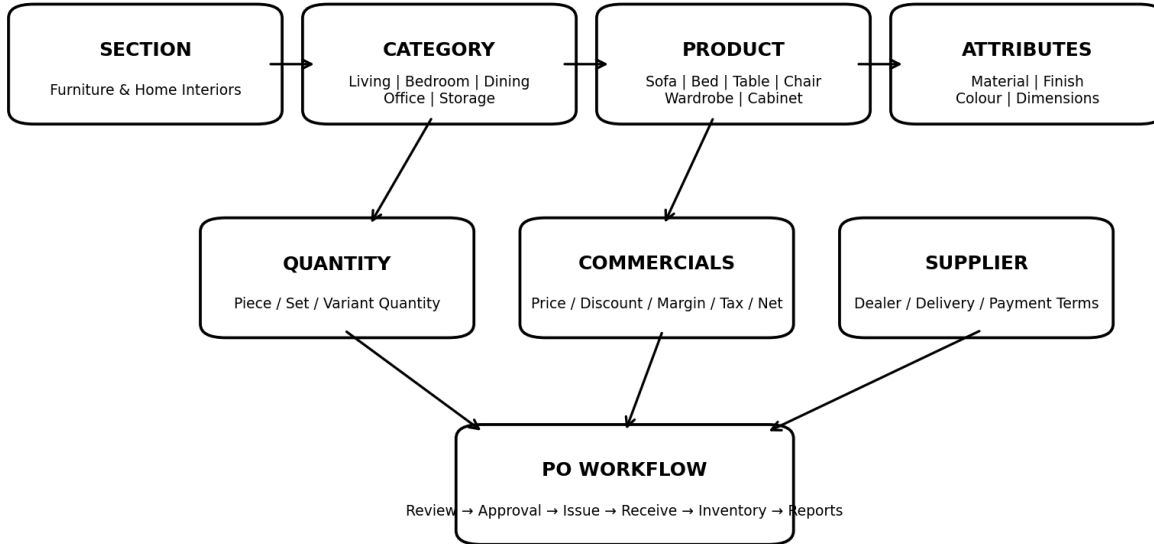


Figure 10. Furniture & Home Interiors procurement flow from section configuration to PO, receipt, inventory and reporting.

Business Area	Included in Application	Example Products / Subcategories	Special Data Structure
Men's Wear	Yes	Shirts, T-shirts, trousers, jeans, ethnic wear, jackets, blazers, innerwear, accessories	S-XXXL / waist sizes / colour × size
Women's Wear	Yes	Kurtis, salwar sets, western wear, innerwear, blouses, accessories	S-XXXL / configurable
Sarees	Yes	Silk, chiffon, satin, tissue, jamdani, banarasi, brocade and other configurable sarees	Free size / fabric / pattern / colour / length
Jewellery	Yes	Necklaces, chains, earrings, bangles, bracelets, rings, pendants, sets	Free size / adjustable / numeric sizes / material / weight
Kids Wear	Yes	Boys, girls, infant, toddler, ethnic, western, footwear, accessories	Age group / size / colour
Toys	Yes	Educational, soft toys, ride-ons, dolls/action figures, games, baby toys	Age group / material / safety / quantity

PURCHASE ORDER & MERCHANDISE PROCUREMENT MANAGEMENT SYSTEM

Business Area	Included in Application	Example Products / Subcategories	Special Data Structure
Accessories	Yes	Belts, bags, watches, wallets, fashion accessories	Free size / model / colour
Home Furnishing	Yes	Bedsheets, blankets, curtains, towels, carpets, mats, cushions, table/sofa covers	Dimensions / material / pattern / colour / unit
Furniture & Home Interiors	Added	Sofas, beds, wardrobes, tables, dining sets, chairs, office furniture, cabinets, TV units, outdoor furniture and custom furniture	Dimensions / material / finish / assembly / warranty / delivery lead time